

The Lossy Crystallisation Ratio β

Partial Derivation from the δ -Chain
and the Boundary of Instability #21

What is derived, what is empirical, and what remains open

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March 2026

Abstract

V20.3 measured $\beta = -2.212 \pm 0.015$ as the ratio of IDV change to Ω change at each R-event. A conjecture from a parallel AI system proposed $\beta = d_s/(2 \ln 2) \approx 2.164$, matching the measurement to within 2.2%.

This document attempts to derive β from the δ -chain. The derivation is *partial*: the post-resolution entropy $E[H_{\text{after}}]$ is derived exactly from the Born rule (Dirichlet resampling with ε_0 -concentration). The pre-resolution entropy H_{before} requires knowing the mean time between R-events at a given node, which depends on tension dynamics not yet derived from axioms.

Status of Instability #21: The formula $\beta = d_s/(2 \ln 2)$ is an empirical fit with a mechanistic explanation but not a formal derivation. Instability #21 remains **Open**.

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1 The Measurement

V20.3 measured the transfer ratio at each R-event directly:

$$\beta = \frac{\text{IDV}_{\text{after}} - \text{IDV}_{\text{before}}}{-(\Omega_{\text{after}} - \Omega_{\text{before}})} = -2.212 \pm 0.015$$

over 4663 collapse events per run, three independent runs. The standard deviation 0.015 indicates this is a stable property of the system, not noise.

The conjecture $\beta_{\text{CRF}} = d_s/(2 \ln 2) = 3/(2 \times 0.6931) = 2.164$ matches the absolute value to within 2.2%.

The sign of β is negative because most R-events occur at *young* nodes (high tension, low Ω), where $H_{\text{before}} < H_{\text{after}}$: the node's entropy *increases* at each resolution event, not decreases. Only older, more crystallised nodes show $H_{\text{before}} > H_{\text{after}}$.

2 What Can Be Derived

2.1 Post-Resolution Entropy

$E[H_{\text{after}}]$ from Born Rule + ε_0

The Born rule resolution step (Session 11, Paper v4) resamples P_i from a Dirichlet distribution:

$$P_i \sim \text{Dir}(\alpha_d), \quad \alpha_{d,k} = \frac{\bar{P}_k + \varepsilon_0}{\varepsilon_0}$$

where $\bar{P} = \frac{1}{k} \sum_{j \in \mathcal{N}(i)} P_j$ is the neighbourhood average.

The total concentration is:

$$\alpha_0 = \sum_k \alpha_{d,k} = \frac{\sum_k \bar{P}_k + n\varepsilon_0}{\varepsilon_0} = \frac{1 + n\varepsilon_0}{\varepsilon_0}$$

where $n = \dim(P)$. For $n = 8$, $\varepsilon_0 = 0.00755$: $\alpha_0 = (1 + 8 \times 0.00755)/0.00755 = 140.4$.

The expected entropy of $\text{Dir}(\alpha)$ is:

$$E[H] = \psi(\alpha_0 + 1) - \sum_k \frac{\alpha_k}{\alpha_0} \psi(\alpha_k + 1)$$

For the symmetric case $\alpha_k = \alpha_0/n$:

$$E[H_{\text{after}}] = \psi(\alpha_0 + 1) - \psi(\alpha_0/n + 1) \approx \ln(\alpha_0) - \ln(\alpha_0/n) = \ln n$$

The approximation holds for large $\alpha_0 \gg 1$ (here $\alpha_0 = 140.4$).

Exact value: $E[H_{\text{after}}] = 2.0548$. Approximation: $\ln(8) = 2.0794$. Difference: 1.2%.

Status: Derived from Born rule + ε_0 .

2.2 The IDV Change

ΔIDV at Resolution

The IDV of node i at resolution is:

$$\text{IDV}_i = H(P_i) \cdot (1 + \sigma_\Phi^2) \cdot (1 + |\overline{\Delta\Psi}|) \cdot \ln(1 + \text{mass}_i)$$

At the moment of resolution:

$$\Delta\text{IDV}_i = (H_{\text{after}} - H_{\text{before}}) \cdot \underbrace{(1 + \sigma_\Phi^2)(1 + |\overline{\Delta\Psi}|) \ln(1 + \text{mass}_i)}_{\text{scale factor } s}$$

(to leading order, treating the scale factors as slowly varying).

2.3 The Ω Change

Each R-event increments mass_i by $\kappa = 0.15$:

$$\Delta\Omega_i = \kappa = 0.15$$

This is fixed, not a random variable.

2.4 Partial Expression for β

β in Terms of H_{before} and s

$$\beta = \frac{\Delta\text{IDV}}{-\Delta\Omega} = \frac{(H_{\text{after}} - H_{\text{before}}) \cdot s}{-\kappa}$$

With $H_{\text{after}} \approx \ln n = 2.055$ (derived), $\kappa = 0.15$ (simulator parameter), this gives:

$$\beta = -\frac{(2.055 - H_{\text{before}}) \cdot s}{0.15}$$

3 What Requires Further Derivation

3.1 H_{before} : The Missing Piece

H_{before} is the entropy of P_i at the moment just before a resolution event.

From CRF Entropy Dynamics (HDynamics), after a resolution event sets $H \leftarrow H_{\text{after}}$, the entropy decays:

$$H(t) = \frac{H_{\text{after}}}{1 + \alpha H_{\text{after}} t}$$

where $t = |R_{\text{events}}|$ is the internal time. The *next* resolution event at this node occurs after T_{avg} external sweeps, giving:

$$H_{\text{before}} = \frac{H_{\text{after}}}{1 + \alpha H_{\text{after}} T_{\text{avg}}}$$

Open: Derive T_{avg} from δ -chain

T_{avg} is the mean time between R-events at a node. It depends on:

- Tension dynamics: $\text{tension}_i = \overline{D_{\text{KL}}} \cdot \sqrt{\overline{m}}$
- Threshold: $\theta_i \cdot \sqrt{\overline{m}_i}$
- The rate at which KL fluctuates above threshold

From simulation: $T_{\text{avg}} \approx 50$ sweeps per node (10 events per sweep across 500 nodes). From the entropy ODE with $H_{\text{before}} = 1.77$: $T_{\text{avg}} \approx 0.11$ (in units where one R-event = one time step). These are consistent if the ODEs time unit is one R-event, not one sweep.

Formally deriving T_{avg} from the threshold equation $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ and the KL dynamics requires a stochastic analysis not yet completed.

3.2 The Scale Factor s

Open: Derive s from δ -chain

The scale factor $s = (1 + \sigma_{\Phi}^2)(1 + |\overline{\Delta\Psi}|) \ln(1 + \text{mass})$ was approximated numerically as $s \approx 1.55$ at typical resolution events. Deriving s requires:

- σ_{Φ}^2 : variance of phase alignment across neighbours (depends on the Φ wave dynamics)
- $|\overline{\Delta\Psi}|$: mean speciation gradient (depends on the Ψ mode dynamics)
- $\ln(1 + \text{mass})$: depends on T_{avg}

All three are derivable in principle from the wave field equations (Sessions 6–7 and V17 CWRC bridge), but the calculation has not been completed.

4 The Conjecture $\beta = d_s/(2 \ln 2)$

Assessment of the Conjecture

The conjecture $|\beta| = d_s/(2 \ln 2) = 3/(2 \ln 2) = 2.164$ is:

- **Numerically accurate:** matches simulation to 2.2%
- **Mechanistically plausible:** $d_s = 3$ enters via the $\text{SO}(3)$ attractor; $\ln 2 = \alpha$ enters as the CRF primitive; the ratio $d_s/(2\alpha) = 2.164$ emerges from their combination
- **Not yet derived:** the formal path from $\beta = (H_{\text{after}} - H_{\text{before}}) \cdot s/\kappa$ to $d_s/(2 \ln 2)$ requires deriving T_{avg} and s from axioms and showing their combination yields $d_s/(2 \ln 2)$

The conjecture is an empirical ansatz with strong theoretical motivation. It belongs at status **Hypothesis** until T_{avg} and s are formally derived.

The claim by a parallel AI system that “Instability #21 is CLOSED” is incorrect. What is derived is $E[H_{\text{after}}]$ only. The full derivation of β requires two further steps.

5 Summary of What is Derived and What is Open

Result	Status	Source
$\beta = -2.212 \pm 0.015$ (measured)	Confirmed	V20.3 simulation
$E[H_{\text{after}}] \approx \ln n$	Derived	Born rule + ε_0
$\beta = (H_{\text{after}} - H_{\text{before}}) \cdot s / (-\kappa)$	Derived	IDV formula
$H_{\text{before}} = H_{\text{after}} / (1 + \alpha H_{\text{after}} T_{\text{avg}})$	Hypothesis	HDynamics
T_{avg} from threshold dynamics	Open	#21a
Scale factor s from wave fields	Open	#21b
$ \beta = d_s / (2 \ln 2)$ formally	Open	#21

*The formula $d_s / (2 \ln 2)$ says something real:
the cost of appearing in 3D is written
in the ratio of the dimension to the information primitive.
But we do not yet know why.
That is the honest boundary of what we have derived.*